In[ ]:=
```
n1 = 9;
g1 = 3;
n2 = 9;
g2 = 1;
n3 = 9;
g3 = 1;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.40093784733070004;
β12 = 0.20631963826157407;
β22bar = 0.8025057114461955;

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5318181818181819; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -0.9489464140282534;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in the middle region. This is obtained by solving two
  consistency conditions along with other constant γ12bar. *)
Q2 = -1.989094374486371`;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
 Q2 f3^(3/2)/Sqrt[x] in the inner region. This is obtained by solving three
    consistency conditions along with other constants β13 & β23. *)
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
R = 3 a/(8 * alfas) * (3 π η/2)^(2/3);(*Page number 54, it is inverse in the copy though *)
c = 12/(5 π) 4/3 g^2 4/3 alfas/a (3 π η/2)^(1/3);
(*Multiplication factor in the expression of kinetic energy *)
d = 4/3 g^2 4/3 alfas/a (3 π η/2)^(1/3) 18/(25 π) η/(η - 1/5);  (* 18g^2/(5(5η -1 )).16 π^2.(3a/8alfas). R^2/(6π^2)^2 ,
The value of R is from page number 54 which is written inverse here,
and (3a/8alfas) is inverse in copy, BA, 13 September *)
(*Multiplication factor in the expression of Potential energy *)


(*Let us define the three functions for three regions *)

f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
```

```
(*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(* s2 is the first derivative of function f_2(x) *)
(*u2 is the function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
  (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
    (* It means at the boundary x3 the function df1 changes to other function *)
    y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
    (*It means at the boundary x3 the function f1 changes to other function*)
   }, (* It means at the boundary x3 the function f1 changes to other function *)
   y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
   condition of (d/dx) f1(x)|x2 = γ12bar (d/dx) f2(x)|x2 and f1(x3)=γ12 f2(x3)  *)
```

```
f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3/2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2/(3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).   *)

(*Start the program*)

m2 = 50;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
```

```
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]) / x^(1/2),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
```

```
solution1 = NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3 , y1[x3] == γ12bar * u3 }, y1, {x, x3, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)

normmx2 = x2 s2 - u2 ;
normx2 = n2 Q2 / (3 η) ;
(*normx2barr and normx2bar should be equal *)
normmx3 = x3 s3 - u3 ;
normx3 = n3 Q3 / (3 η) + β22bar n2 Q2 / (3 η) - (β22bar)^(3/2) n2 Q3 / (3 η);
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
normx1 = n1 Q1 / (3 η) + γ12bar n3 Q3 / (3 η) - (γ12bar)^(3/2) n3 Q1 / (3 η) +
  ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) n2 Q1 / (3 η) - γ12bar (β22bar)^(3/2) n2 Q3 / (3 η) + γ12bar β22bar n2 Q2 / (3 η) ;
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;

AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;

CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;

CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)


K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;

K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;

(*K2bar2barr and K2bar2bar should be equal *)

K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2}-\frac{\beta22bar}{Q3}+\frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}s3\,;$$

$$K332=\frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-\frac{n2\,\beta22bar}{3\,\eta\,Q3}s2+\frac{n2}{3\,\eta\,Q3}s3-\frac{(\beta22bar)^{\frac{3}{2}}}{Q2}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\beta12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\beta12^{1.5}+\frac{\beta12^{1.5}}{Q2}-\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2+$$

$$\frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3}-\frac{\gamma12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}(\beta12)^{\frac{3}{2}}s2+\frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}s3-$$

$$\frac{n2}{3\,\eta\,Q3}(\gamma12bar)^{\frac{3}{2}}\beta22bar\,s2+\frac{n2}{3\,\eta\,Q1}s1-\frac{n2}{3\,\eta\,Q1}\gamma12bar\,s3-\frac{(\beta12)^{\frac{3}{2}}}{Q2}h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}-\frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\gamma12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$

$$\frac{n2}{3\,\eta}\beta22bar\left(\frac{5}{7\,Q2}\beta22bar^{\frac{1}{2}}*\beta12^{1.5}-\frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\gamma12bar^{1.5}*\right.$$

$$\left.\beta22bar+\frac{Q2}{Q3^2}\gamma12bar^{1.5}*\beta22bar+\frac{1}{Q2}\beta12^{1.5}*\beta22bar^{\frac{1}{2}}-\frac{\beta12^{1.5}}{Q3}\right)s2+$$

$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5}-\frac{n3\,Q3}{Q1}\gamma12bar+n2*\beta12^{1.5}-\frac{n2\,Q2}{Q3}\gamma12bar^{1.5}*\beta22bar+\right.$$

$$\left.\frac{5}{7}\gamma12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta22bar-n2*\beta22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} - \frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) + \frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} + \frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar - \frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) + \frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 - \frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) * s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
    Print["x2 to x3 (h5 computationally)=  ", h5];
    Print["x2 to x3 (h5 by hand)=  ", CD];
    Print["x2 to x3 (h5 by handdddd)=  ", CCD];
    Print["x3 to x1 (h6 computationally)=  ", h6];
    Print["x3 to x1 (h6 by hand)=  ", EF];
    Print["K22  (computationally) = ", K222];
    Print["K22  (by hand) = ", K22];
    Print["K32  (computationally) = ", K332];
    Print["K32  (by hand) = ", K32];
    Print["K21  (computationally) = ", K221];
    Print["K21  (by hand) =", K21];
    Print["K31  (computationally) = ", K331];
    Print["K31  (by hand) = ", K31];
    Print["K33  (computationally) = ", K333];
    Print["K33  (by hand) = ", K33];
    Print["K11  (computationally) = ", K111];
    Print["K11  (by hand) = ", K11];
    Print["x2= ", x2];
    Print["x3= ", x3];
    Print["x1=  ", x1];
     Print["T by hand=", T1];
    Print["V by hand=", V1];
    Print["m2=", m2];
    Print["R=  ", R];
Print["Energywithoutmass = ", Energywithoutmass];
    Print["Energy=  ", Energy];
    If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
      Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
       H = 10^9 + 1;
       m2 = m2B;
       If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
   If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
   If[n == 3, Energy6bare = HX;
    Print["x1=", x1];
    Print["x3=", x3];
    Print["x2=", x2];
    Print["Kinetic Energy=", T1];
    Print["Potential Energy=", V1];
    Print["Volume Energy=", Volumeenergy];
   ]]];

(*Code for Plotting function *)
```

```
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.043074757
normx2 (computationally)=  -0.663031306
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000866
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.1954026
0 to x2 (h4 by hand)=  16.1954012
x2 to x3 (h5 computationally)=  3.5610582
x2 to x3 (h5 by hand)=  3.56106307
x2 to x3 (h5 by handdddd)=  3.5610599
x3 to x1 (h6 computationally)=  4.09030549
x3 to x1 (h6 by hand)=  4.0903062
K22  (computationally) = 0.933193356
K22  (by hand) = 0.93319263
K32  (computationally) = 0.876018963
K32  (by hand) = 0.876018441
K21  (computationally) = 0.566096518
K21  (by hand) =0.56609645
K31  (computationally) = 0.560463865
K31  (by hand) = 0.560465118
K33  (computationally) = 0.833834557
K33  (by hand) = 0.833839309
K11  (computationally) = 0.489932035
K11  (by hand) = 0.489933705
x2= 0.143478646
x3= 0.160744114
x1=  0.26673229
T by hand=1202.46955
V by hand=-232.299226
m2=50
R=  0.043074757
```

```
Energywithoutmass = 1821.7651
Energy=  38 037.7651
R=  0.043074757
normx2 (computationally)=  −0.663031306
normx2 (by hand)=  −0.663031458
normx3 (computationally)=  −0.621000824
normx3 (by hand)=  −0.621000866
normx1 (computationally)=  −0.396991763
normx1 (by hand)=  −0.396991764
0 to x2 (h4 computationally)=  16.1986891
0 to x2 (h4 by hand)=  16.1986876
x2 to x3 (h5 computationally)=  3.56178878
x2 to x3 (h5 by hand)=  3.56179257
x2 to x3 (h5 by handdddd)=  3.56179047
x3 to x1 (h6 computationally)=  4.09116359
x3 to x1 (h6 by hand)=  4.09116487
K22  (computationally) = 0.933287647
K22  (by hand) = 0.93328692
K32  (computationally) = 0.876107514
K32  (by hand) = 0.876106992
K21  (computationally) = 0.566154705
K21  (by hand) =0.566154637
K31  (computationally) = 0.560521079
K31  (by hand) = 0.560522046
K33  (computationally) = 0.833917272
K33  (by hand) = 0.833920896
K11  (computationally) = 0.489984122
K11  (by hand) = 0.489986784
x2= 0.143464087
x3= 0.160727734
x1=  0.266704521
T by hand=1202.71893
V by hand=−232.323331
m2=50.01
R=  0.043074757
Energywithoutmass = 1821.72443
```

```
Energy=  38 037.7244
R=  0.043074757
normx2 (computationally)=  −0.663031396
normx2 (by hand)=  −0.663031458
normx3 (computationally)=  −0.621000852
normx3 (by hand)=  −0.621000866
normx1 (computationally)=  −0.396991752
normx1 (by hand)=  −0.396991764
0 to x2 (h4 computationally)=  16.2282696
0 to x2 (h4 by hand)=  16.2282696
x2 to x3 (h5 computationally)=  3.56837726
x2 to x3 (h5 by hand)=  3.5683799
x2 to x3 (h5 by handdddd)=  3.56837896
x3 to x1 (h6 computationally)=  4.0988782
x3 to x1 (h6 by hand)=  4.09887908
K22  (computationally) = 0.934137043
K22  (by hand) = 0.934137023
K32  (computationally) = 0.876905663
K32  (by hand) = 0.876905649
K21  (computationally) = 0.566677611
K21  (by hand) =0.566677609
K31  (computationally) = 0.561038522
K31  (by hand) = 0.561039226
K33  (computationally) = 0.834675977
K33  (by hand) = 0.834678746
K11  (computationally) = 0.490438345
K11  (by hand) = 0.490440184
x2= 0.143333266
x3= 0.160580606
x1=  0.266454868
T by hand=1204.96158
V by hand=−232.537333
m2=50.1
R=  0.043074757
Energywithoutmass = 1821.3646
Energy=  38 037.3646
```

```
R=  0.043074757
normx2 (computationally)=  -0.66303134
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000866
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.2611335
0 to x2 (h4 by hand)=  16.2611326
x2 to x3 (h5 computationally)=  3.57570233
x2 to x3 (h5 by hand)=  3.57570651
x2 to x3 (h5 by handdddd)=  3.57570404
x3 to x1 (h6 computationally)=  4.10745077
x3 to x1 (h6 by hand)=  4.10745149
K22  (computationally) = 0.935077839
K22  (by hand) = 0.935077381
K32  (computationally) = 0.877790408
K32  (by hand) = 0.877790079
K21  (computationally) = 0.567257959
K21  (by hand) =0.567257916
K31  (computationally) = 0.561613918
K31  (by hand) = 0.561615005
K33  (computationally) = 0.835521609
K33  (by hand) = 0.835525775
K11  (computationally) = 0.490943576
K11  (by hand) = 0.490945203
x2= 0.143188314
x3= 0.16041761
x1=  0.266178296
T by hand=1207.4537
V by hand=-232.77532
m2=50.2
R=  0.043074757
Energywithoutmass = 1820.97795
Energy=  38 036.978
R=  0.043074757
```

```
normx2 (computationally)=  -0.663031406
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.62100084
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991757
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.2940008
0 to x2 (h4 by hand)=  16.2940009
x2 to x3 (h5 computationally)=  3.58302207
x2 to x3 (h5 by hand)=  3.58302421
x2 to x3 (h5 by handdddd)=  3.58302378
x3 to x1 (h6 computationally)=  4.11602451
x3 to x1 (h6 by hand)=  4.11602557
K22  (computationally) = 0.936019315
K22  (by hand) = 0.936019367
K32  (computationally) = 0.878675121
K32  (by hand) = 0.878675159
K21  (computationally) = 0.567837889
K21  (by hand) =0.567837893
K31  (computationally) = 0.562187499
K31  (by hand) = 0.562188075
K33  (computationally) = 0.836361461
K33  (by hand) = 0.836363743
K11  (computationally) = 0.491449373
K11  (by hand) = 0.491451511
x2= 0.143043807
x3= 0.160255089
x1=  0.26590258
T by hand=1209.94596
V by hand=-233.013083
m2=50.3
R=  0.043074757
Energywithoutmass = 1820.60531
Energy=  38 036.6053
R=  0.043074757
normx2 (computationally)=  -0.663031405
```

```
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000824
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991764
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.326867
0 to x2 (h4 by hand)=  16.3268671
x2 to x3 (h5 computationally)=  3.59034452
x2 to x3 (h5 by hand)=  3.59034625
x2 to x3 (h5 by handdddd)=  3.59034624
x3 to x1 (h6 computationally)=  4.12459848
x3 to x1 (h6 by hand)=  4.12459978
K22  (computationally) = 0.936958983
K22  (by hand) = 0.936959035
K32  (computationally) = 0.879558472
K32  (by hand) = 0.879558509
K21  (computationally) = 0.568417174
K21  (by hand) =0.568417179
K31  (computationally) = 0.562761081
K31  (by hand) = 0.562761548
K33  (computationally) = 0.837202648
K33  (by hand) = 0.8372045
K11  (computationally) = 0.491954554
K11  (by hand) = 0.491957128
x2= 0.142899731
x3= 0.160093066
x1=  0.265627709
T by hand=1212.43834
V by hand=-233.250694
m2=50.4
R=  0.043074757
Energywithoutmass = 1820.2464
Energy=  38 036.2464
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.043074757
normx2 (computationally)=  -0.663031357
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000853
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.3597326
0 to x2 (h4 by hand)=  16.3597319
x2 to x3 (h5 computationally)=  3.59767055
x2 to x3 (h5 by hand)=  3.59767406
x2 to x3 (h5 by handdddd)=  3.59767227
x3 to x1 (h6 computationally)=  4.13317169
x3 to x1 (h6 by hand)=  4.13317251
K22  (computationally) = 0.937897085
K22  (by hand) = 0.937896762
K32  (computationally) = 0.880440678
K32  (by hand) = 0.880440446
K21  (computationally) = 0.568995761
K21  (by hand) =0.568995731
K31  (computationally) = 0.563334823
K31  (by hand) = 0.56333574
K33  (computationally) = 0.838046219
K33  (by hand) = 0.838049749
K11  (computationally) = 0.492457404
K11  (by hand) = 0.492459182
x2= 0.142756083
x3= 0.159931542
x1=  0.265353664
T by hand=1214.93059
V by hand=-233.487769
m2=50.5
R=  0.043074757
Energywithoutmass = 1819.90112
Energy=  38 035.9011
R=  0.043074757
```

```
normx2 (computationally)=  -0.663031405
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.62100085
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991763
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.3926008
0 to x2 (h4 by hand)=  16.3926009
x2 to x3 (h5 computationally)=  3.60499235
x2 to x3 (h5 by hand)=  3.60499477
x2 to x3 (h5 by handdddd)=  3.60499408
x3 to x1 (h6 computationally)=  4.14174648
x3 to x1 (h6 by hand)=  4.14174759
K22  (computationally) = 0.938835535
K22  (by hand) = 0.938835587
K32  (computationally) = 0.88132268
K32  (by hand) = 0.881322718
K21  (computationally) = 0.569573951
K21  (by hand) =0.569573956
K31  (computationally) = 0.563907053
K31  (by hand) = 0.563907705
K33  (computationally) = 0.838885002
K33  (by hand) = 0.838887585
K11  (computationally) = 0.492961678
K11  (by hand) = 0.49296393
x2= 0.142612873
x3= 0.15977049
x1=  0.26508047
T by hand=1217.42314
V by hand=-233.724909
m2=50.6
R=  0.043074757
Energywithoutmass = 1819.56951
Energy=  38 035.5695
R=  0.043074757
normx2 (computationally)=  -0.663031328
```

```
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000829
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991749
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.4254665
0 to x2 (h4 by hand)=  16.4254654
x2 to x3 (h5 computationally)=  3.61231772
x2 to x3 (h5 by hand)=  3.61232126
x2 to x3 (h5 by handdddd)=  3.61231946
x3 to x1 (h6 computationally)=  4.15032104
x3 to x1 (h6 by hand)=  4.15032205
K22  (computationally) = 0.939771382
K22  (by hand) = 0.939770825
K32  (computationally) = 0.8822028
K32  (by hand) = 0.8822024
K21  (computationally) = 0.570151399
K21  (by hand) =0.570151347
K31  (computationally) = 0.564479413
K31  (by hand) = 0.564480322
K33  (computationally) = 0.839725589
K33  (by hand) = 0.839729033
K11  (computationally) = 0.493464481
K11  (by hand) = 0.493466616
x2= 0.142470082
x3= 0.159609928
x1=  0.2648081
T by hand=1219.91566
V by hand=-233.961667
m2=50.7
R=  0.043074757
Energywithoutmass = 1819.25134
Energy=  38 035.2513
R=  0.043074757
normx2 (computationally)=  -0.663031364
normx2 (by hand)=  -0.663031458
```

```
normx3 (computationally)=  -0.621000837
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.39699175
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.4583355
0 to x2 (h4 by hand)=  16.4583349
x2 to x3 (h5 computationally)=  3.61964072
x2 to x3 (h5 by hand)=  3.6196437
x2 to x3 (h5 by handdddd)=  3.61964246
x3 to x1 (h6 computationally)=  4.1588961
x3 to x1 (h6 by hand)=  4.15889708
K22  (computationally) = 0.940707822
K22  (by hand) = 0.940707548
K32  (computationally) = 0.883082993
K32  (by hand) = 0.883082796
K21  (computationally) = 0.570728398
K21  (by hand) =0.570728372
K31  (computationally) = 0.565050667
K31  (by hand) = 0.565051445
K33  (computationally) = 0.84056348
K33  (by hand) = 0.840566478
K11  (computationally) = 0.493966927
K11  (by hand) = 0.493968987
x2= 0.142327725
x3= 0.159449841
x1=  0.264536559
T by hand=1222.40827
V by hand=-234.198159
m2=50.8
R=  0.043074757
Energywithoutmass = 1818.94663
Energy=  38 034.9466
R=  0.043074757
normx2 (computationally)=  -0.663031404
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000823
```

```
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991763
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.491205
0 to x2 (h4 by hand)=  16.4912051
x2 to x3 (h5 computationally)=  3.62696291
x2 to x3 (h5 by hand)=  3.62696469
x2 to x3 (h5 by handdddd)=  3.62696466
x3 to x1 (h6 computationally)=  4.16747207
x3 to x1 (h6 by hand)=  4.16747339
K22  (computationally) = 0.941643409
K22  (by hand) = 0.941643457
K32  (computationally) = 0.883962314
K32  (by hand) = 0.883962348
K21  (computationally) = 0.571304872
K21  (by hand) =0.571304876
K31  (computationally) = 0.565621151
K31  (by hand) = 0.56562163
K33  (computationally) = 0.841399531
K33  (by hand) = 0.841401428
K11  (computationally) = 0.494469793
K11  (by hand) = 0.494472394
x2= 0.142185792
x3= 0.159290229
x1=  0.264265849
T by hand=1224.90107
V by hand=-234.434599
m2=50.9
R=  0.043074757
Energywithoutmass = 1818.65524
Energy=  38 034.6552
R=  0.043074757
normx2 (computationally)=  -0.663031337
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000829
normx3 (by hand)=  -0.621000866
```

```
normx1 (computationally)=  -0.39699175
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.5240723
0 to x2 (h4 by hand)=  16.5240713
x2 to x3 (h5 computationally)=  3.63428959
x2 to x3 (h5 by hand)=  3.63429295
x2 to x3 (h5 by handdddd)=  3.63429135
x3 to x1 (h6 computationally)=  4.17604749
x3 to x1 (h6 by hand)=  4.17604854
K22  (computationally) = 0.942576621
K22  (by hand) = 0.942576134
K32  (computationally) = 0.884839967
K32  (by hand) = 0.884839617
K21  (computationally) = 0.57188061
K21  (by hand) =0.571880564
K31  (computationally) = 0.566191998
K31  (by hand) = 0.566192864
K33  (computationally) = 0.842238484
K33  (by hand) = 0.842241773
K11  (computationally) = 0.494970728
K11  (by hand) = 0.494972933
x2= 0.142044273
x3= 0.159131104
x1=  0.263995948
T by hand=1227.39381
V by hand=-234.670617
m2=51.
R=  0.043074757
Energywithoutmass = 1818.37699
Energy=  38 034.377
R=  0.043074757
normx2 (computationally)=  -0.663031404
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000823
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991758
```

```
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.5569433
0 to x2 (h4 by hand)=  16.5569434
x2 to x3 (h5 computationally)=  3.64161161
x2 to x3 (h5 by hand)=  3.6416134
x2 to x3 (h5 by handdddd)=  3.64161337
x3 to x1 (h6 computationally)=  4.18462398
x3 to x1 (h6 by hand)=  4.18462521
K22  (computationally) = 0.943510734
K22  (by hand) = 0.943510782
K32  (computationally) = 0.88571779
K32  (by hand) = 0.885717825
K21  (computationally) = 0.572455958
K21  (by hand) =0.572455962
K31  (computationally) = 0.566761224
K31  (by hand) = 0.566761706
K33  (computationally) = 0.843072502
K33  (by hand) = 0.843074413
K11  (computationally) = 0.495472157
K11  (by hand) = 0.495474599
x2= 0.141903184
x3= 0.15897244
x1=  0.263726868
T by hand=1229.88671
V by hand=-234.906458
m2=51.1
R=  0.043074757
Energywithoutmass = 1818.11197
Energy=  38 034.112
R=  0.043074757
normx2 (computationally)=  -0.663031404
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000826
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991764
normx1 (by hand)=  -0.396991764
```

```
0 to x2 (h4 computationally)=  16.5898132
0 to x2 (h4 by hand)=  16.5898133
x2 to x3 (h5 computationally)=  3.64893634
x2 to x3 (h5 by hand)=  3.64893821
x2 to x3 (h5 by handdddd)=  3.64893811
x3 to x1 (h6 computationally)=  4.19320068
x3 to x1 (h6 by hand)=  4.193202
K22  (computationally) = 0.94444303
K22  (by hand) = 0.944443081
K32  (computationally) = 0.886594248
K32  (by hand) = 0.886594285
K21  (computationally) = 0.573030673
K21  (by hand) =0.573030678
K31  (computationally) = 0.567330465
K31  (by hand) = 0.567330967
K33  (computationally) = 0.843907871
K33  (by hand) = 0.84390986
K11  (computationally) = 0.495972979
K11  (by hand) = 0.495975583
x2= 0.141762508
x3= 0.158814253
x1=  0.2634586
T by hand=1232.37971
V by hand=-235.142151
m2=51.2
R=  0.043074757
Energywithoutmass = 1817.85992
Energy=  38 033.8599
R=  0.043074757
normx2 (computationally)=  -0.663031314
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000823
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991764
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.6226813
```

```
0 to x2 (h4 by hand)=  16.62268
x2 to x3 (h5 computationally)=  3.65626402
x2 to x3 (h5 by hand)=  3.65626774
x2 to x3 (h5 by handdddd)=  3.65626579
x3 to x1 (h6 computationally)=  4.20177766
x3 to x1 (h6 by hand)=  4.201779
K22  (computationally) = 0.945373195
K22  (by hand) = 0.94537253
K32  (computationally) = 0.887469129
K32  (by hand) = 0.887468651
K21  (computationally) = 0.573604735
K21  (by hand) =0.573604673
K31  (computationally) = 0.567899761
K31  (by hand) = 0.567900712
K33  (computationally) = 0.844744641
K33  (by hand) = 0.844748219
K11  (computationally) = 0.496473286
K11  (by hand) = 0.496476056
x2= 0.141622241
x3= 0.158656542
x1=  0.26319114
T by hand=1234.87283
V by hand=-235.377712
m2=51.3
R=  0.043074757
Energywithoutmass = 1817.62075
Energy=  38033.6207
R=  0.043074757
normx2 (computationally)=  -0.663031365
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000826
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991756
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.6555533
0 to x2 (h4 by hand)=  16.6555528
```

```
x2 to x3 (h5 computationally)=  3.66358747
x2 to x3 (h5 by hand)=  3.66359019
x2 to x3 (h5 by handdddd)=  3.66358925
x3 to x1 (h6 computationally)=  4.21035473
x3 to x1 (h6 by hand)=  4.21035592
K22  (computationally) = 0.946304352
K22  (by hand) = 0.946304089
K32  (computationally) = 0.888344272
K32  (by hand) = 0.888344083
K21  (computationally) = 0.574178341
K21  (by hand) =0.574178316
K31  (computationally) = 0.568467481
K31  (by hand) = 0.56846819
K33  (computationally) = 0.845577006
K33  (by hand) = 0.845579733
K11  (computationally) = 0.49697234
K11  (by hand) = 0.496974753
x2= 0.141482397
x3= 0.158499286
x1=  0.262924474
T by hand=1237.36585
V by hand=-235.612662
m2=51.4
R=  0.043074757
Energywithoutmass = 1817.39456
Energy=  38033.3946
R=  0.043074757
normx2 (computationally)=  -0.663031365
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000823
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.39699175
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.6884246
0 to x2 (h4 by hand)=  16.6884241
x2 to x3 (h5 computationally)=  3.67091255
```

```
x2 to x3 (h5 by hand)=  3.6709152
x2 to x3 (h5 by handdddd)=  3.67091434
x3 to x1 (h6 computationally)=  4.21893228
x3 to x1 (h6 by hand)=  4.21893339
K22  (computationally) = 0.947233927
K22  (by hand) = 0.947233668
K32  (computationally) = 0.889218159
K32  (by hand) = 0.889217973
K21  (computationally) = 0.574751351
K21  (by hand) =0.574751327
K31  (computationally) = 0.569034977
K31  (by hand) = 0.569035667
K33  (computationally) = 0.846409699
K33  (by hand) = 0.846412352
K11  (computationally) = 0.497471195
K11  (by hand) = 0.497473452
x2= 0.141342962
x3= 0.158342497
x1=  0.262658611
T by hand=1239.85901
V by hand=-235.847513
m2=51.5
R=  0.043074757
Energywithoutmass = 1817.1811
Energy=  38 033.1811
R=  0.043074757
normx2 (computationally)=  -0.66303137
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000847
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991752
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.7212964
0 to x2 (h4 by hand)=  16.7212959
x2 to x3 (h5 computationally)=  3.67823878
x2 to x3 (h5 by hand)=  3.67824197
```

```
x2 to x3 (h5 by handdddd)=  3.67824057
x3 to x1 (h6 computationally)=  4.22751012
x3 to x1 (h6 by hand)=  4.22751108
K22  (computationally) = 0.948162659
K22  (by hand) = 0.948162439
K32  (computationally) = 0.890091292
K32  (by hand) = 0.890091134
K21  (computationally) = 0.575323811
K21  (by hand) =0.57532379
K31  (computationally) = 0.569602167
K31  (by hand) = 0.569603005
K33  (computationally) = 0.847242627
K33  (by hand) = 0.847245872
K11  (computationally) = 0.497969516
K11  (by hand) = 0.497971553
x2= 0.141203936
x3= 0.158186175
x1=  0.262393544
T by hand=1242.35228
V by hand=-236.082197
m2=51.6
R=  0.043074757
Energywithoutmass = 1816.98029
Energy=  38 032.9803
R=  0.043074757
normx2 (computationally)=  -0.663031384
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.62100084
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991752
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.7541688
0 to x2 (h4 by hand)=  16.7541686
x2 to x3 (h5 computationally)=  3.68556359
x2 to x3 (h5 by hand)=  3.68556629
x2 to x3 (h5 by handdddd)=  3.68556539
```

x3 to x1 (h6 computationally)= 4.23608859

x3 to x1 (h6 by hand)= 4.2360896

K22 (computationally) = 0.949090616

K22 (by hand) = 0.949090508

K32 (computationally) = 0.890963581

K32 (by hand) = 0.890963503

K21 (computationally) = 0.575895735

K21 (by hand) =0.575895725

K31 (computationally) = 0.570168426

K31 (by hand) = 0.570169141

K33 (computationally) = 0.848073127

K33 (by hand) = 0.848075915

K11 (computationally) = 0.49846772

K11 (by hand) = 0.49846981

x2= 0.141065318

x3= 0.158030305

x1= 0.26212927

T by hand=1244.84564

V by hand=−236.316646

m2=51.7

R= 0.043074757

Energywithoutmass = 1816.79211

Energy= 38 032.7921

R= 0.043074757

normx2 (computationally)= −0.663031302

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000823

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991753

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 16.7870394

0 to x2 (h4 by hand)= 16.7870378

x2 to x3 (h5 computationally)= 3.69289139

x2 to x3 (h5 by hand)= 3.69289543

x2 to x3 (h5 by handdddd)= 3.69289319

x3 to x1 (h6 computationally)= 4.2446676

```
x3 to x1 (h6 by hand)=  4.24466878
K22  (computationally) = 0.950016366
K22  (by hand) = 0.950015598
K32  (computationally) = 0.891834244
K32  (by hand) = 0.891833692
K21  (computationally) = 0.576467031
K21  (by hand) =0.576466959
K31  (computationally) = 0.570734778
K31  (by hand) = 0.570735806
K33  (computationally) = 0.848905051
K33  (by hand) = 0.848908905
K11  (computationally) = 0.498965921
K11  (by hand) = 0.498968399
x2= 0.140927099
x3= 0.157874899
x1=  0.261865788
T by hand=1247.33919
V by hand=-236.551101
m2=51.8
R=  0.043074757
Energywithoutmass = 1816.61635
Energy=  38 032.6164
R=  0.043074757
normx2 (computationally)=  -0.663031404
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000849
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.8199149
0 to x2 (h4 by hand)=  16.819915
x2 to x3 (h5 computationally)=  3.70021503
x2 to x3 (h5 by hand)=  3.70021756
x2 to x3 (h5 by handdddd)=  3.70021684
x3 to x1 (h6 computationally)=  4.25324629
x3 to x1 (h6 by hand)=  4.25324717
```

K22 (computationally) = 0.950943737

K22 (by hand) = 0.950943792

K32 (computationally) = 0.892705625

K32 (by hand) = 0.892705664

K21 (computationally) = 0.577037897

K21 (by hand) =0.577037903

K31 (computationally) = 0.571299586

K31 (by hand) = 0.571300267

K33 (computationally) = 0.849732983

K33 (by hand) = 0.849735677

K11 (computationally) = 0.499462157

K11 (by hand) = 0.499463987

x2= 0.140789298

x3= 0.157719941

x1= 0.261603076

T by hand=1249.83257

V by hand=-236.784805

m2=51.9

R= 0.043074757

Energywithoutmass = 1816.45316

Energy= 38 032.4532

R= 0.043074757

normx2 (computationally)= -0.663031371

normx2 (by hand)= -0.663031458

normx3 (computationally)= -0.621000847

normx3 (by hand)= -0.621000866

normx1 (computationally)= -0.396991757

normx1 (by hand)= -0.396991764

0 to x2 (h4 computationally)= 16.8527875

0 to x2 (h4 by hand)= 16.8527871

x2 to x3 (h5 computationally)= 3.70754218

x2 to x3 (h5 by hand)= 3.70754537

x2 to x3 (h5 by handdddd)= 3.70754399

x3 to x1 (h6 computationally)= 4.26182607

x3 to x1 (h6 by hand)= 4.26182714

K22 (computationally) = 0.951868372

```
K22  (by hand) = 0.951868163
K32  (computationally) = 0.893575032
K32  (by hand) = 0.893574881
K21  (computationally) = 0.577608144
K21  (by hand) =0.577608124
K31  (computationally) = 0.571864634
K31  (by hand) = 0.571865474
K33  (computationally) = 0.850562597
K33  (by hand) = 0.850565855
K11  (computationally) = 0.499959442
K11  (by hand) = 0.499961666
x2= 0.140651888
x3= 0.157565441
x1=  0.261341158
T by hand=1252.32628
V by hand=-237.018797
m2=52.
R=  0.043074757
Energywithoutmass = 1816.30218
Energy=  38 032.3022
R=  0.043074757
normx2 (computationally)=  -0.6630314
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000823
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991749
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.8856622
0 to x2 (h4 by hand)=  16.8856622
x2 to x3 (h5 computationally)=  3.71486656
x2 to x3 (h5 by hand)=  3.7148685
x2 to x3 (h5 by handdddd)=  3.71486838
x3 to x1 (h6 computationally)=  4.27040597
x3 to x1 (h6 by hand)=  4.27040708
K22  (computationally) = 0.952792961
K22  (by hand) = 0.952792985
```

K32 (computationally) = 0.894444044

K32 (by hand) = 0.894444062

K21 (computationally) = 0.578177869

K21 (by hand) =0.578177871

K31 (computationally) = 0.572428417

K31 (by hand) = 0.572428938

K33 (computationally) = 0.851388765

K33 (by hand) = 0.851390825

K11 (computationally) = 0.500455595

K11 (by hand) = 0.500457806

x2= 0.140514882

x3= 0.157411383

x1= 0.261080008

T by hand=1254.81992

V by hand=−237.252237

m2=52.1

R= 0.043074757

Energywithoutmass = 1816.16355

Energy= 38 032.1635

R= 0.043074757

normx2 (computationally)= −0.663031403

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000866

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991748

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 16.9185366

0 to x2 (h4 by hand)= 16.9185367

x2 to x3 (h5 computationally)= 3.72219472

x2 to x3 (h5 by hand)= 3.72219774

x2 to x3 (h5 by handdddd)= 3.72219655

x3 to x1 (h6 computationally)= 4.27898566

x3 to x1 (h6 by hand)= 4.27898642

K22 (computationally) = 0.953716315

K22 (by hand) = 0.953716366

K32 (computationally) = 0.895312169

```
K32  (by hand) = 0.895312206
K21  (computationally) = 0.578746999
K21  (by hand) =0.578747003
K31  (computationally) = 0.572992514
K31  (by hand) = 0.572993324
K33  (computationally) = 0.852217749
K33  (by hand) = 0.852220953
K11  (computationally) = 0.500950518
K11  (by hand) = 0.500952154
x2= 0.140378272
x3= 0.157257787
x1=  0.260819626
T by hand=1257.31362
V by hand=-237.485483
m2=52.2
R=  0.043074757
Energywithoutmass = 1816.037
Energy=  38 032.037
R=  0.043074757
normx2 (computationally)=  -0.663031301
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000866
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  16.9514088
0 to x2 (h4 by hand)=  16.9514072
x2 to x3 (h5 computationally)=  3.72952487
x2 to x3 (h5 by hand)=  3.72953013
x2 to x3 (h5 by handdddd)=  3.7295267
x3 to x1 (h6 computationally)=  4.28756622
x3 to x1 (h6 by hand)=  4.28756698
K22  (computationally) = 0.954637329
K22  (by hand) = 0.954636548
K32  (computationally) = 0.89617852
K32  (by hand) = 0.896177959
```

K21 (computationally) = 0.579315515

K21 (by hand) =0.579315442

K31 (computationally) = 0.573556471

K31 (by hand) = 0.573557828

K33 (computationally) = 0.853047022

K33 (by hand) = 0.85305217

K11 (computationally) = 0.501445975

K11 (by hand) = 0.501447755

x2= 0.14024205

x3= 0.157104639

x1= 0.260560022

T by hand=1259.80755

V by hand=−237.718798

m2=52.3

R= 0.043074757

Energywithoutmass = 1815.92248

Energy= 38 031.9225

R= 0.043074757

normx2 (computationally)= −0.66303132

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000853

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991751

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 16.9842845

0 to x2 (h4 by hand)= 16.9842832

x2 to x3 (h5 computationally)= 3.7368506

x2 to x3 (h5 by hand)= 3.73685513

x2 to x3 (h5 by handdddd)= 3.73685244

x3 to x1 (h6 computationally)= 4.2961474

x3 to x1 (h6 by hand)= 4.29614833

K22 (computationally) = 0.955559125

K22 (by hand) = 0.955558496

K32 (computationally) = 0.897044982

K32 (by hand) = 0.89704453

K21 (computationally) = 0.579883614

```
K21  (by hand) =0.579883555
K31  (computationally) = 0.574118845
K31  (by hand) = 0.574120015
K33  (computationally) = 0.853871579
K33  (by hand) = 0.853876031
K11  (computationally) = 0.501941141
K11  (by hand) = 0.501943182
x2= 0.140106231
x3= 0.156951922
x1=  0.260301181
T by hand=1262.30151
V by hand=-237.951733
m2=52.4
R=  0.043074757
Energywithoutmass = 1815.82007
Energy=  38031.8201
R=  0.043074757
normx2 (computationally)=  -0.663031319
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000849
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991757
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.0171601
0 to x2 (h4 by hand)=  17.0171588
x2 to x3 (h5 computationally)=  3.74417751
x2 to x3 (h5 by hand)=  3.74418196
x2 to x3 (h5 by handdddd)=  3.74417936
x3 to x1 (h6 computationally)=  4.30472891
x3 to x1 (h6 by hand)=  4.30472997
K22  (computationally) = 0.95647977
K22  (by hand) = 0.95647913
K32  (computationally) = 0.897910477
K32  (by hand) = 0.897910017
K21  (computationally) = 0.580451152
K21  (by hand) =0.580451092
```

K31 (computationally) = 0.574680917

K31 (by hand) = 0.574682064

K33 (computationally) = 0.854696253

K33 (by hand) = 0.854700611

K11 (computationally) = 0.502435845

K11 (by hand) = 0.502438149

x2= 0.139970802

x3= 0.156799651

x1= 0.260043102

T by hand=1264.79558

V by hand=−238.184501

m2=52.5

R= 0.043074757

Energywithoutmass = 1815.72956

Energy= 38 031.7296

R= 0.043074757

normx2 (computationally)= −0.663031404

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000841

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991761

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 17.0500383

0 to x2 (h4 by hand)= 17.0500384

x2 to x3 (h5 computationally)= 3.75150154

x2 to x3 (h5 by hand)= 3.7515039

x2 to x3 (h5 by handdddd)= 3.75150339

x3 to x1 (h6 computationally)= 4.31331075

x3 to x1 (h6 by hand)= 4.31331195

K22 (computationally) = 0.957400734

K22 (by hand) = 0.957400791

K32 (computationally) = 0.898775836

K32 (by hand) = 0.898775877

K21 (computationally) = 0.581018226

K21 (by hand) =0.581018231

K31 (computationally) = 0.575241745

```
K31  (by hand) = 0.575242381
K33  (computationally) = 0.85551756
K33  (by hand) = 0.85552008
K11  (computationally) = 0.502929859
K11  (by hand) = 0.502932275
x2= 0.13983577
x3= 0.156647812
x1=  0.259785777
T by hand=1267.28964
V by hand=-238.416857
m2=52.6
R=  0.043074757
Energywithoutmass = 1815.65098
Energy=  38 031.651
R=  0.043074757
normx2 (computationally)=  -0.663031372
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000833
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991753
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.082914
0 to x2 (h4 by hand)=  17.0829136
x2 to x3 (h5 computationally)=  3.75882969
x2 to x3 (h5 by hand)=  3.75883253
x2 to x3 (h5 by handdddd)=  3.75883154
x3 to x1 (h6 computationally)=  4.32189261
x3 to x1 (h6 by hand)=  4.32189373
K22  (computationally) = 0.958319213
K22  (by hand) = 0.958319014
K32  (computationally) = 0.899639435
K32  (by hand) = 0.899639293
K21  (computationally) = 0.581584625
K21  (by hand) =0.581584607
K31  (computationally) = 0.575802911
K31  (by hand) = 0.575803658
```

```
K33  (computationally) = 0.856341351
K33  (by hand) = 0.856344244
K11  (computationally) = 0.503423018
K11  (by hand) = 0.503425313
x2= 0.139701118
x3= 0.15649642
x1=  0.259529203
T by hand=1269.78378
V by hand=-238.649049
m2=52.7
R=  0.043074757
Energywithoutmass = 1815.58407
Energy=  38 031.5841
R=  0.043074757
normx2 (computationally)=  -0.6630313
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000851
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991753
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.1157892
0 to x2 (h4 by hand)=  17.1157876
x2 to x3 (h5 computationally)=  3.76616052
x2 to x3 (h5 by hand)=  3.76616547
x2 to x3 (h5 by handdddd)=  3.76616238
x3 to x1 (h6 computationally)=  4.33047475
x3 to x1 (h6 by hand)=  4.33047573
K22  (computationally) = 0.959236248
K22  (by hand) = 0.959235456
K32  (computationally) = 0.900501941
K32  (by hand) = 0.900501371
K21  (computationally) = 0.582150452
K21  (by hand) =0.582150377
K31  (computationally) = 0.576364142
K31  (by hand) = 0.576365412
K33  (computationally) = 0.857166686
```

```
K33  (by hand) = 0.857171488
K11  (computationally) = 0.503915757
K11  (by hand) = 0.503917937
x2= 0.13956685
x3= 0.156345471
x1=  0.25927338
T by hand=1272.27806
V by hand=-238.881171
m2=52.8
R=  0.043074757
Energywithoutmass = 1815.52878
Energy=  38 031.5288
R=  0.043074757
normx2 (computationally)=  -0.6630313
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000822
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991751
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.1486665
0 to x2 (h4 by hand)=  17.148665
x2 to x3 (h5 computationally)=  3.77348711
x2 to x3 (h5 by hand)=  3.77349128
x2 to x3 (h5 by handdddd)=  3.77348898
x3 to x1 (h6 computationally)=  4.33905772
x3 to x1 (h6 by hand)=  4.33905889
K22  (computationally) = 0.960153419
K22  (by hand) = 0.960152628
K32  (computationally) = 0.90136411
K32  (by hand) = 0.901363541
K21  (computationally) = 0.582715831
K21  (by hand) =0.582715757
K31  (computationally) = 0.576923812
K31  (by hand) = 0.576924876
K33  (computationally) = 0.857987124
K33  (by hand) = 0.857991118
```

K11 (computationally) = 0.504408582

K11 (by hand) = 0.504411052

x2= 0.139432971

x3= 0.156194942

x1= 0.259018306

T by hand=1274.77241

V by hand=−239.11298

m2=52.9

R= 0.043074757

Energywithoutmass = 1815.4852

Energy= 38 031.4852

R= 0.043074757

normx2 (computationally)= −0.663031302

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000834

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991751

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 17.1815444

0 to x2 (h4 by hand)= 17.1815428

x2 to x3 (h5 computationally)= 3.78081551

x2 to x3 (h5 by hand)= 3.78081997

x2 to x3 (h5 by handdddd)= 3.78081738

x3 to x1 (h6 computationally)= 4.34764065

x3 to x1 (h6 by hand)= 4.34764175

K22 (computationally) = 0.961069746

K22 (by hand) = 0.961068968

K32 (computationally) = 0.902225566

K32 (by hand) = 0.902225007

K21 (computationally) = 0.58328066

K21 (by hand) =0.583280587

K31 (computationally) = 0.577483343

K31 (by hand) = 0.577484484

K33 (computationally) = 0.858808523

K33 (by hand) = 0.858812824

K11 (computationally) = 0.504900385

```
K11  (by hand) = 0.504902737
x2= 0.139299475
x3= 0.156044852
x1=  0.258763972
T by hand=1277.2668
V by hand=-239.344539
m2=53.
R=  0.043074757
Energywithoutmass = 1815.45311
Energy=  38 031.4531
R=  0.043074757
normx2 (computationally)=  -0.6630313
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000832
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.2144226
0 to x2 (h4 by hand)=  17.214421
x2 to x3 (h5 computationally)=  3.78814367
x2 to x3 (h5 by hand)=  3.78814815
x2 to x3 (h5 by handdddd)=  3.78814555
x3 to x1 (h6 computationally)=  4.35622398
x3 to x1 (h6 by hand)=  4.35622502
K22  (computationally) = 0.961985153
K22  (by hand) = 0.961984356
K32  (computationally) = 0.903086149
K32  (by hand) = 0.903085576
K21  (computationally) = 0.583844949
K21  (by hand) =0.583844874
K31  (computationally) = 0.578042237
K31  (by hand) = 0.578043381
K33  (computationally) = 0.859628677
K33  (by hand) = 0.859632985
K11  (computationally) = 0.505391746
K11  (by hand) = 0.505394011
```

```
x2= 0.13916636
x3= 0.155895189
x1=  0.258510375
T by hand=1279.76126
V by hand=-239.57585
m2=53.1
R=  0.043074757
Energywithoutmass = 1815.43249
Energy=  38 031.4325
R=  0.043074757
normx2 (computationally)=  -0.663031403
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000827
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991753
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.2473039
0 to x2 (h4 by hand)=  17.247304
x2 to x3 (h5 computationally)=  3.79546827
x2 to x3 (h5 by hand)=  3.79547033
x2 to x3 (h5 by handdddd)=  3.79547016
x3 to x1 (h6 computationally)=  4.36480794
x3 to x1 (h6 by hand)=  4.36480912
K22  (computationally) = 0.96290117
K22  (by hand) = 0.962901225
K32  (computationally) = 0.903946771
K32  (by hand) = 0.903946811
K21  (computationally) = 0.584408827
K21  (by hand) =0.584408832
K31  (computationally) = 0.578599768
K31  (by hand) = 0.578600324
K33  (computationally) = 0.860444927
K33  (by hand) = 0.860447127
K11  (computationally) = 0.505882954
K11  (by hand) = 0.505885323
x2= 0.139033631
```

```
x3= 0.155745944
x1=  0.258257516
T by hand=1282.25579
V by hand=-239.80687
m2=53.2
R=  0.043074757
Energywithoutmass = 1815.42332
Energy=  38 031.4233
R=  0.043074757
normx2 (computationally)=  -0.663031329
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000851
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991759
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.2801811
0 to x2 (h4 by hand)=  17.28018
x2 to x3 (h5 computationally)=  3.80280032
x2 to x3 (h5 by hand)=  3.80280469
x2 to x3 (h5 by handdddd)=  3.80280221
x3 to x1 (h6 computationally)=  4.37339191
x3 to x1 (h6 by hand)=  4.37339301
K22  (computationally) = 0.963813857
K22  (by hand) = 0.963813302
K32  (computationally) = 0.904805211
K32  (by hand) = 0.904804812
K21  (computationally) = 0.584971993
K21  (by hand) =0.584971941
K31  (computationally) = 0.579158444
K31  (by hand) = 0.579159574
K33  (computationally) = 0.861266703
K33  (by hand) = 0.861271016
K11  (computationally) = 0.506373552
K11  (by hand) = 0.506375921
x2= 0.138901268
x3= 0.155597144
```

```
x1=  0.25800539
T by hand=1284.75049
V by hand=-240.03797
m2=53.3
R=  0.043074757
Energywithoutmass = 1815.42527
Energy=  38 031.4253
R=  0.043074757
normx2 (computationally)=  -0.663031402
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000846
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.2505918
0 to x2 (h4 by hand)=  17.2505919
x2 to x3 (h5 computationally)=  3.79620192
x2 to x3 (h5 by hand)=  3.7962045
x2 to x3 (h5 by handdddd)=  3.79620381
x3 to x1 (h6 computationally)=  4.36566595
x3 to x1 (h6 by hand)=  4.36566689
K22  (computationally) = 0.962992578
K22  (by hand) = 0.962992632
K32  (computationally) = 0.904032748
K32  (by hand) = 0.904032786
K21  (computationally) = 0.584465149
K21  (by hand) =0.584465154
K31  (computationally) = 0.578655753
K31  (by hand) = 0.578656447
K33  (computationally) = 0.860527673
K33  (by hand) = 0.860530421
K11  (computationally) = 0.5059314
K11  (by hand) = 0.505933337
x2= 0.139020379
x3= 0.155731047
x1=  0.258232266
```

```
T by hand=1282.50517
V by hand=-239.829858
m2=53.21
R=  0.043074757
Energywithoutmass = 1815.42301
Energy=  38 031.423
R=  0.043074757
normx2 (computationally)=  -0.663031299
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.621000845
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.253877
0 to x2 (h4 by hand)=  17.2538754
x2 to x3 (h5 computationally)=  3.79693824
x2 to x3 (h5 by hand)=  3.79694308
x2 to x3 (h5 by handdddd)=  3.79694013
x3 to x1 (h6 computationally)=  4.36652436
x3 to x1 (h6 by hand)=  4.36652531
K22  (computationally) = 0.963082534
K22  (by hand) = 0.963081731
K32  (computationally) = 0.904117828
K32  (by hand) = 0.904117251
K21  (computationally) = 0.584521396
K21  (by hand) =0.584521321
K31  (computationally) = 0.578712349
K31  (by hand) = 0.578713591
K33  (computationally) = 0.860612423
K33  (by hand) = 0.860617113
K11  (computationally) = 0.505980704
K11  (by hand) = 0.505982807
x2= 0.139007122
x3= 0.155716159
x1=  0.25820703
T by hand=1282.75472
```

V by hand=−239.853161

m2=53.22

R= 0.043074757

Energywithoutmass = 1815.42273

Energy= 38 031.4227

R= 0.043074757

normx2 (computationally)= −0.663031331

normx2 (by hand)= −0.663031458

normx3 (computationally)= −0.621000852

normx3 (by hand)= −0.621000866

normx1 (computationally)= −0.396991748

normx1 (by hand)= −0.396991764

0 to x2 (h4 computationally)= 17.2571657

0 to x2 (h4 by hand)= 17.2571647

x2 to x3 (h5 computationally)= 3.79767031

x2 to x3 (h5 by hand)= 3.79767465

x2 to x3 (h5 by handdddd)= 3.7976722

x3 to x1 (h6 computationally)= 4.36738267

x3 to x1 (h6 by hand)= 4.36738356

K22 (computationally) = 0.963174374

K22 (by hand) = 0.963173836

K32 (computationally) = 0.904204024

K32 (by hand) = 0.904203637

K21 (computationally) = 0.584577758

K21 (by hand) =0.584577708

K31 (computationally) = 0.578767971

K31 (by hand) = 0.578769095

K33 (computationally) = 0.860693733

K33 (by hand) = 0.860698025

K11 (computationally) = 0.506029577

K11 (by hand) = 0.506031539

x2= 0.138993879

x3= 0.155701266

x1= 0.258181798

T by hand=1283.00414

V by hand=−239.876182

```
m2=53.23
R=  0.043074757
Energywithoutmass = 1815.42268
Energy=  38 031.4227
R=  0.043074757
normx2 (computationally)=  -0.663031402
normx2 (by hand)=  -0.663031458
normx3 (computationally)=  -0.62100084
normx3 (by hand)=  -0.621000866
normx1 (computationally)=  -0.396991748
normx1 (by hand)=  -0.396991764
0 to x2 (h4 computationally)=  17.2604554
0 to x2 (h4 by hand)=  17.2604555
x2 to x3 (h5 computationally)=  3.79840024
x2 to x3 (h5 by hand)=  3.79840268
x2 to x3 (h5 by handdddd)=  3.79840213
x3 to x1 (h6 computationally)=  4.36824122
x3 to x1 (h6 by hand)=  4.3682422
K22  (computationally) = 0.963266752
K22  (by hand) = 0.963266802
K32  (computationally) = 0.904290488
K32  (by hand) = 0.904290524
K21  (computationally) = 0.584634163
K21  (by hand) =0.584634168
K31  (computationally) = 0.578823094
K31  (by hand) = 0.578823751
K33  (computationally) = 0.860773078
K33  (by hand) = 0.860775677
K11  (computationally) = 0.506078711
K11  (by hand) = 0.506080721
x2= 0.138980643
x3= 0.155686371
x1=  0.258156573
T by hand=1283.25357
V by hand=-239.89916
m2=53.24
```

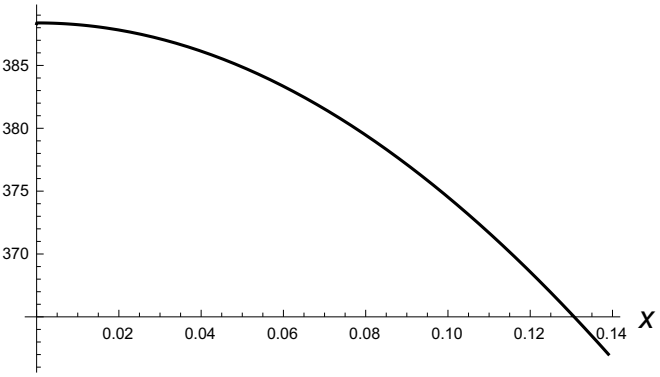
```
R=  0.043074757
Energywithoutmass = 1815.42279
Energy=  38 031.4228
R=  0.043074757
normx2 (computationally)=  −0.663031366
normx2 (by hand)=  −0.663031458
normx3 (computationally)=  −0.621000854
normx3 (by hand)=  −0.621000866
normx1 (computationally)=  −0.396991752
normx1 (by hand)=  −0.396991764
0 to x2 (h4 computationally)=  17.2574954
0 to x2 (h4 by hand)=  17.2574949
x2 to x3 (h5 computationally)=  3.79774248
x2 to x3 (h5 by hand)=  3.79774609
x2 to x3 (h5 by handdddd)=  3.79774437
x3 to x1 (h6 computationally)=  4.36746861
x3 to x1 (h6 by hand)=  4.36746956
K22  (computationally) = 0.963183998
K22  (by hand) = 0.963183748
K32  (computationally) = 0.904212902
K32  (by hand) = 0.904212722
K21  (computationally) = 0.584583432
K21  (by hand) =0.584583409
K31  (computationally) = 0.578773306
K31  (by hand) = 0.578774255
K33  (computationally) = 0.860701012
K33  (by hand) = 0.860704687
K11  (computationally) = 0.506034599
K11  (by hand) = 0.506036628
x2= 0.138992557
x3= 0.155699775
x1=  0.258179276
T by hand=1283.02909
V by hand=−239.878479
m2=53.231
R=  0.043074757
```

```
Energywithoutmass = 1815.4227
Energy=  38 031.4227
R=  0.043074757
normx2 (computationally)=  −0.663031366
normx2 (by hand)=  −0.663031458
normx3 (computationally)=  −0.621000854
normx3 (by hand)=  −0.621000866
normx1 (computationally)=  −0.39699175
normx1 (by hand)=  −0.396991764
0 to x2 (h4 computationally)=  17.2578242
0 to x2 (h4 by hand)=  17.2578237
x2 to x3 (h5 computationally)=  3.79781577
x2 to x3 (h5 by hand)=  3.79781939
x2 to x3 (h5 by handdddd)=  3.79781766
x3 to x1 (h6 computationally)=  4.3675544
x3 to x1 (h6 by hand)=  4.36755532
K22  (computationally) = 0.963193135
K22  (by hand) = 0.963192883
K32  (computationally) = 0.904221493
K32  (by hand) = 0.904221312
K21  (computationally) = 0.584589062
K21  (by hand) =0.584589038
K31  (computationally) = 0.578778884
K31  (by hand) = 0.578779834
K33  (computationally) = 0.860709209
K33  (by hand) = 0.860712889
K11  (computationally) = 0.506039416
K11  (by hand) = 0.506041387
x2= 0.138991233
x3= 0.155698286
x1=  0.258176752
T by hand=1283.05403
V by hand=−239.880765
m2=53.232
R=  0.043074757
Energywithoutmass = 1815.4227
```

```
Energy=  38 031.4227
x1=0.258176752
x3=0.155698286
x2=0.138991233
Kinetic Energy=1283.05403
Potential Energy=-239.880765
Volume Energy=772.249443
```

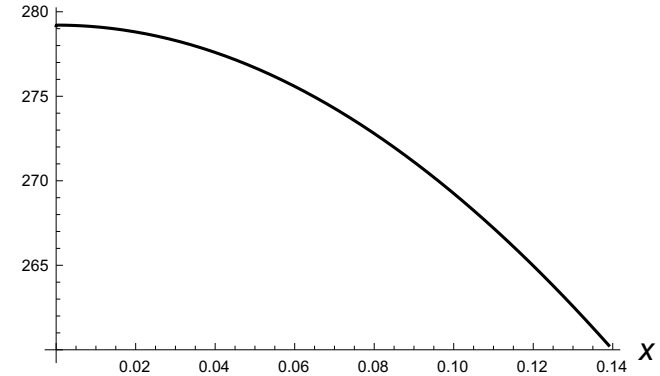
$(f_2(x)/x)^{3/2}$

Out[ ]=



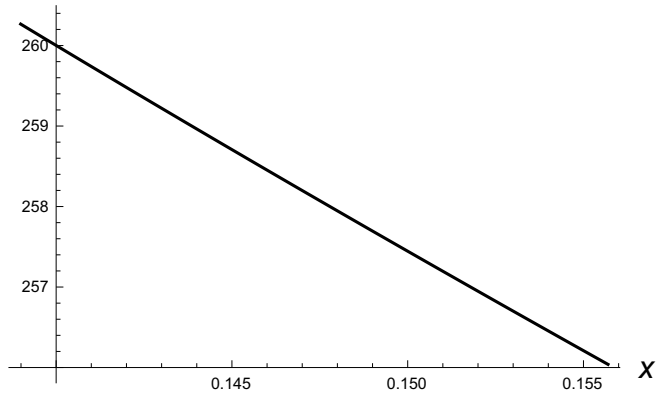
$(f_{2\,bar}(x)/x)^{3/2}$

Out[ ]=

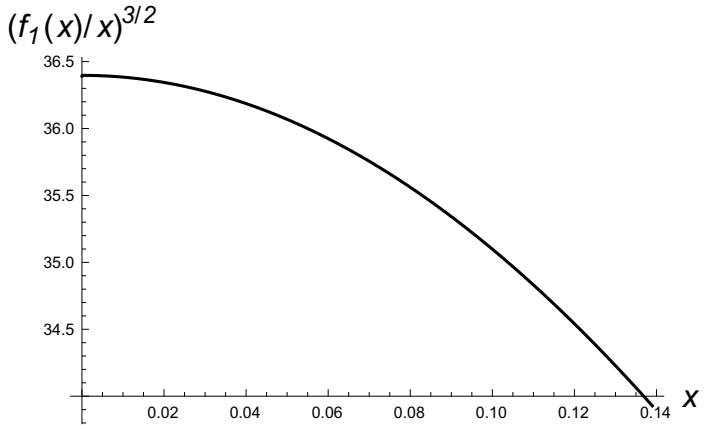


$(f_{2\,bar}(x)/x)^{3/2}$

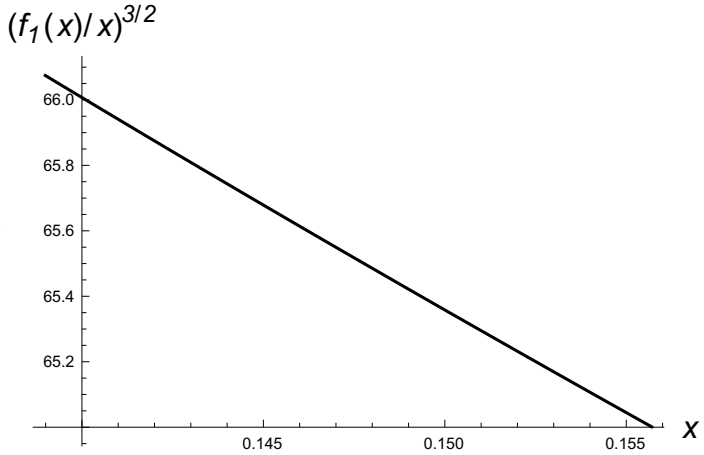
Out[ ]=



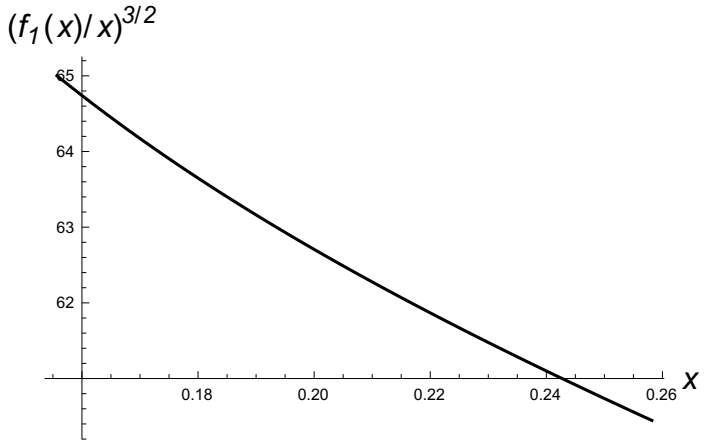
Out[ ]=


$(f_1(x)/x)^{3/2}$
36.5
36.0
35.5
35.0
34.5
0.02
0.04
0.06
0.08
0.10
0.12
0.14
x

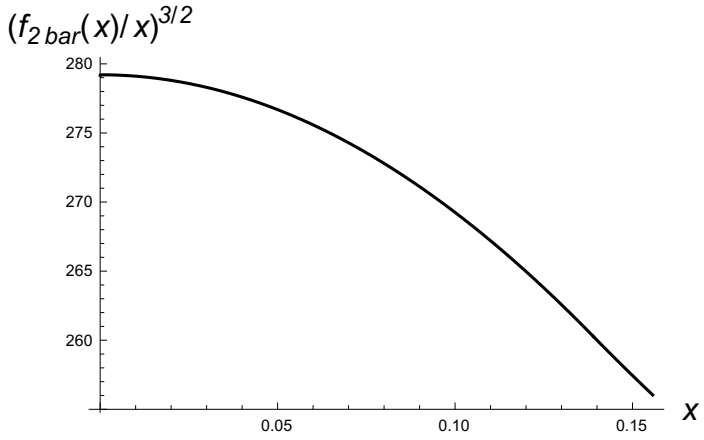
Out[ ]=


$(f_1(x)/x)^{3/2}$
66.0
65.8
65.6
65.4
65.2
0.145
0.150
0.155
x

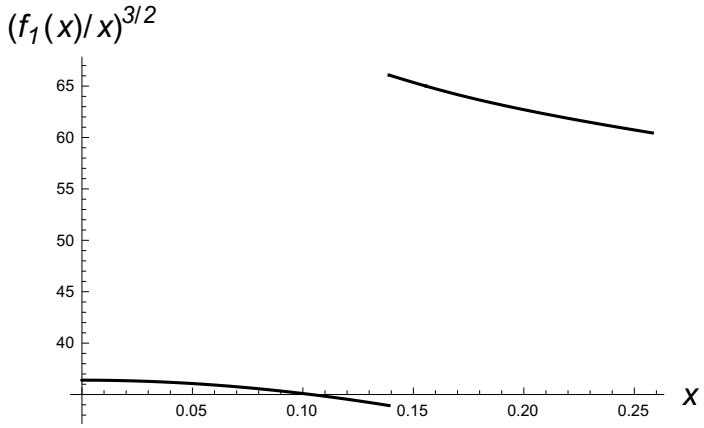
Out[ ]=


$(f_1(x)/x)^{3/2}$
65
64
63
62
0.18
0.20
0.22
0.24
0.26
x

Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
280
275
270
265
260
0.05
0.10
0.15
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
65
60
55
50
45
40
0.05
0.10
0.15
0.20
0.25
x